ScholarWatch

Project Team

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Chapter 1

Introduction

Learning Management Systems (LMS) are software applications designed to facilitate the management, delivery, and tracking of educational courses, training programs, or learning and development programs. They provide a centralized platform where instructors can create, manage, and deliver educational content, while learners can access materials, submit assignments, and track their progress.

Some of the most well-known LMS platforms include Canvas, Blackboard, and Moodle.

The intended audience for this document includes:

Developers: Responsible for the technical aspects of Scholarwatch, including model selection for different modules like AI-based quiz generation, AI-based content generation and Engagement monitoring.

Project Managers: Overseeing the project's progress they are interested in the scope, work division, timeline, and module descriptions. Project managers ensure that the Scholarwatch project stays aligned with its objectives and timelines, as outlined in this document.They include supervisor , co-supervisor and the FYP committee.

Instructors: As end-users, instructors will benefit from understanding how Scholarwatch improves their course management process, automates content creation, tracks student attention, and facilitates quiz generation. Their daily operations will rely on the system's effectiveness and reliability.

Students: As end-users, ScholarWatch provides students with an enhanced learning experience by offering personalized content and quizzes that adapt to their individual progress.They can attend lectures from the ease of their home at any time of the day.

Marketing Staff: While not directly involved in the development, marketing staff would focus on Scholarwatch's innovative features, including AI-based student monitoring and smart attendance systems. These aspects can be highlighted to promote adoption within educational institutions..

1.1 Problem Statement

Despite the widespread use of LMS platforms, current systems face several limitations that hinder both teaching and learning experiences.

First, there is a notable lack of real-time attention tracking, preventing teachers from monitoring whether students are engaged and focused during lectures or not.

These platforms also lack features that provide analysis of students' struggling areas, i.e., topics they find difficult to grasp.

Furthermore, the labor of manual slide creation is time-consuming and detracts from educators' focus on more meaningful tasks.

In online learning environments, students often face challenges related to self-discipline and attendance, with no check or balance to ensure that students are attending their classes daily and with concentration.

Additionally, the standardized, one-size-fits-all approach adopted by traditional LMS platforms does not account for students' varying academic abilities. All students are given the same learning experience, which can hinder their academic growth. Scholarwatch, an AI-powered LMS, is specifically designed to address these issues by integrating innovative features that effectively resolve the challenges mentioned above

1.2 Scope

The scope of Scholarwatch involves the optimization of an LMS namely Moodle, that addresses the limitations of traditional educational platforms. The key features of the project work together to create a seamless and personalized learning environment, optimizing traditional systems. These features are listed as follows:

Real-Time Engagement Tracking: Scholarwatch uses various ML models to monitor student focus during lectures, such as gaze tracking, posture, and facial expression detection. These insights identify weak points in student understanding, forming the basis for the next step in the learning process. Gaze, posture, and facial expressions are assessed at different intervals for a specified duration, with the threshold set by the instructor.

Automated Assessment Creation: Based on the insights from engagement tracking, personalized MCQ-based quizzes are generated, targeting each student's weak areas. These quizzes focus on topics where students showed less attention or struggled during the lecture, ensuring assessments not only test but also reinforce learning in those specific areas. The number of MCQs and options is a hyperparameter that can be set by the instructor.

Activity Moinitoring: During these personalized quizzes, Scholarwatch implements anticheating mechanisms such as gaze tracking, tab-switch detection, copy-paste detection, and split-screen monitoring to maintain academic integrity. If cheating is detected, the system can penalize students accordingly, ensuring honesty in assessments.

Post-Assessment Adaptation: Scholarwatch adapts the learning experience by offering tailored and in-depth content to support students who need it most. The instructor can also recommend, based on students' performance, that some take lectures covering a more detailed and simplified version of the topic.

Content Creation Tools: To facilitate the teaching process, Scholarwatch includes tools that automate lecture slide creation. Instructors can either upload their own slides or create new ones using the platform. For generation, the instructor must upload reference material within a specified size limit, clearly mentioning the subtopics or page numbers. The number of slides, styling, and other hyperparameters can be set by the instructor. The generated slides will contain textual data only. User prompts are not entertained; all hyperparameters will be predefined by the instructor through checkboxes. The instructor can also set some lectures as pre-requisites i,e. It is compulsory to finish them before moving towards more advanced topics.

Attendance Monitoring: To ensure engagement, attendance will be monitored based on attentiveness using the models discussed above. The threshold required for a student to be marked present will be set by the instructor.

Insights Analysis: To put all the features into perspective, analytics and insights will be displayed via a dashboard. The instructor will have one dashboard, accompanied by certain filters, while the student will have a separate dashboard focusing solely on their performance based on all the analytics gathered from the implemented features. Together, these components make Scholarwatch a comprehensive solution for enhancing learning outcomes through in-depth customization.

1.3 Modules

1.3.1 User Attention Module

Module focusing on features which primarily require input from the user's webcam.

1. Gaze tracking

2. Posture detection

3. Facial expression recognition

1.3.2 Insights Module

All of student's progress tracking will be worked on in this module, including visualization of the progress.

1. Identification of weak areas of a student
2. Prerequisite tagging of lectures/topics
3. Insights/Analysis of the students

1.3.3 Generation Module

This module consists of models relating to generative tasks, with the help of LLM with RAGs.

1. Content Generation
2. Quiz Generation

1.3.4 Activity Monitoring Module

This module covers areas which require tracking/monitoring students' screen display.

1. Screen Monitoring
2. Tab Switching
3. Detection of any copy pasted content

1.4 User Classes and Characteristics

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User class	Description
Instructors	Instructors are responsible for creating and managing educational content, tracking student progress, and evaluating performance. They would use the system to generate lectures as part of an online lecture, assign quizzes, and monitor student engagement using real-time analytics provided by Scholarwatch. Instructors need an efficient, user-friendly system that allows them to focus on teaching while automating administrative tasks such as quiz generation and attendance tracking..
Students	Students are the primary learners who interact with Scholarwatch to access course materials and take quizzes. They use the system regularly to track their progress and participate in quizzes, which adapt based on their performance while being monitored. Students also benefit from real-time attention tracking to ensure they stay focused during lessons and receive additional support when necessary.
Researchers	Researchers include educational professionals interested in analyzing student engagement data, learning outcomes, and system-generated insights for improving learning methodologies. They might use Scholarwatch occasionally to collect data for research and development in areas such as AI-based education, attention tracking, and personalized learning.

Scholarwatch User Classes and Characteristics

Chapter 2

Project Requirements

This chapter describes the functional and non-functional requirements of the project.

2.1 Use-case/Event Response Table

2.1.1 Use-case

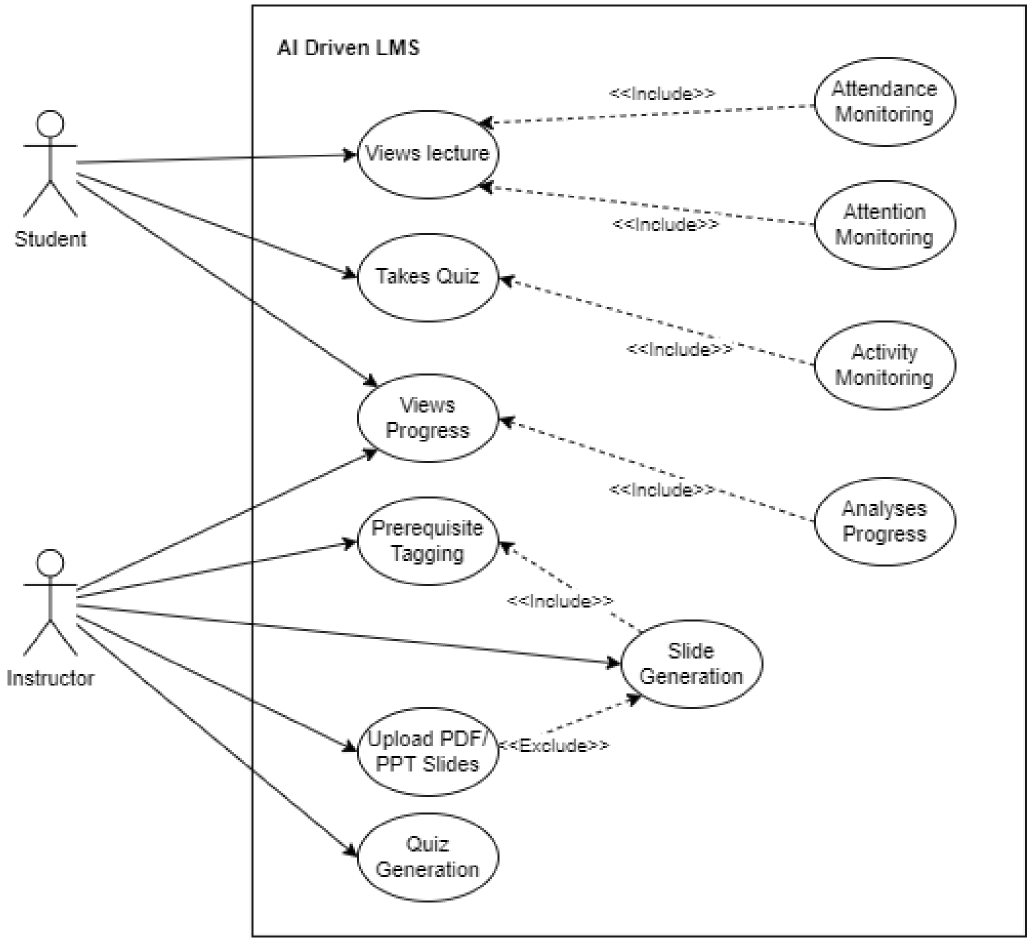

Figure 2.1: Use case diagram

Actors:

Student: A user who interacts with the LMS to attend lectures, take quizzes, and view their progress.

Instructor: A user who manages the course content by tagging prerequisites, generating slides, and quizzes.

Use Cases:

Below is the description of use cases for the **Student**.

1. Attends Lecture
Description: The student views the lecture in the form of a slideshow via the LMS interface.
Includes:

- **Attention Monitoring:** The system monitors the student's engagement by analyzing attention patterns (e.g., through facial emotion recognition, posture, and gaze tracking).

- **Attendance Monitoring:** The system tracks the student's presence during the lecture by periodically assessing their attention at different time intervals.

2. Takes Quiz
Description: The student takes an MCQ-based quiz to assess their knowledge.
Includes:

- **Activity Monitoring:** The system tracks the student's activity, such as tab switching, split-screen usage, and detection of copy-pasted content.

3. Views Progress
Description: The student views their dashboard for a summary of their academic progress and quiz scores.
Includes:

- **Analyzes Progress:** The system analyzes the student's performance in various tasks (e.g., quizzes, attendance) and generates reports on progress, highlighting areas for improvement.

Below are the use cases for the **Instructor**.

4. Views Progress
Description: The instructor reviews the overall progress of the entire class, with detailed records and insights.
Includes:

- **Analyzes Progress:** The system evaluates students' attention across various lecture slides, providing analytics that highlight which slides or topics were more difficult for students to comprehend. Additionally, the system assesses the class's overall quiz and attention performance, offering insights that help the instructor identify areas where students may be struggling and adjust teaching accordingly.

5. Prerequisite Tagging
Description: The instructor tags specific content or concepts as prerequisites for future lessons or tasks, ensuring that students have the necessary foundation.
Includes:

- **Slide Generation:** The system generates relevant slides or content based on the tagged prerequisites and reference material uploaded by the instructor, helping students grasp necessary concepts before advancing.

6. Upload PDF/PPT Slides
Description: The instructor uploads slides or content (in PDF or PPT format) for lectures.
Excludes:

- **Slide Generation:** This task only involves uploading existing slides, without using the system's AI capabilities to generate new content.

7. Quiz Generation
Description: The system creates a quiz for students based on the course material. Hyper-parameters like the number of questions, time limit, and number of options are set by the instructor.

2.1.2 Event Response Table

Event response table

Event	Actor	System Response	Outcome
User signs up	Instructor ,Student	Validate user credentials Provides access based on user role	The user is directed to appropriate interface
User uploads reference material required to generate slides	Instructor	Automatically generates PPT slides and gets feedback for approval	Customized slides ready for slideshow
User tags the lecture as pre-requisites for the upcoming topics	Instructor	Generates knowledge graphs for hierarchical tagging	Hierarchical formatting of content
User attends the lecture via slideshow	Student	Begins monitoring student attention by gaze, emotion and posture detection	Attention data collected and analyzed to detect weak areas
Quizzes are generated	Instructor	Automatically generates quizzes targeting weak areas	MCQ- based graded quizzes
Quiz has started	Student	Activity Monitoring (eye movement, tab switching, split screen detection,copy-paste detection) to prevent cheating	Alerts and penalties are issued for potential cheating detected during quiz
User views performance via dashboards	Student	Display the student's engagement statistics, quiz results, and attention tracking summary	Performance analysis provided to the student
User views students performance via dashboards	Instructor	Display the class's engagement statistics, quiz results, and attention tracking summary	Performance insights provided to teacher

Figure 2.2: Event Response table for ScholarWatch

2.2 Functional Requirements

2.2.1 User Attention Module

Following are the requirements for module 1:

1. The system will use gaze tracking, posture (laying, slouching) and facial expression recognition (yawning, drowsy) to monitor if students are paying attention during lectures.

2. The students' attentiveness will be monitored against an attentiveness threshold set by the instructor to mark students' attendance accordingly

2.2.2 Insights Module

Following are the requirements for module 2:

1. The system will check for gaze periods on specific areas of the slide and screen to identify areas which require extra focus.

2. The instructor will be able to tag lectures as a prerequisite for other lectures/topics.

3. The instructor will be able to view the class' progress with added filters on the dashboard to pinpoint certain students' (top scorers, below average scorers, non-attentive, attendance)

4. The student will also have access to his individual dashboard showcasing his attention span, weak areas, attendance, scores and progress.

2.2.3 Generation Module

Following are the requirements for module 3:

1. The instructor will be given an option to upload a PDF of a chapter/topic to the system, or they can upload their own ppt slides (no changes to be made to these).

2. The instructor will be given PPT slides (textual only) by the system from the uploaded PDF. Check boxes will be given for theme, number of lines per slide and total number of slides.

3. The instructor will be able to tag lectures as a prerequisite for other lectures/topics.

4. The instructor will have the option to create quizzes from the chosen lecture/slides

5. Check boxes will be given to the instructor to specify the number of questions and number of multiple choices to be given to the student in each quiz

2.2.4 Activity Monitoring Module

Following are the requirements for module 4:

1. The system will monitor the students' screen during quizzes.
2. It will detect if a student switches tabs, split screens windows and gaze tracking to identify any potential cheating during assessments.

2.3 Non-Functional Requirements

This section specifies nonfunctional requirements.

2.3.1 Reliability

Scholarwatch must maintain an uptime of atleast 90 % , ensuring 24/7 availability for courses, quizzes, and real-time tracking without interruption.So the students can take their lectures at any time of the day.

2.3.2 Performance

Scholarwatch must process and display real-time student attention data during lectures with minimal latency.Moreover it should ensure quiz generation takes no longer than 10 seconds and grading of quizzes takes up no longer than 5 seconds.

For details on performance and scalability of Moodle, refer to [**?**].

2.3.3 Scalability

The system should support up to 1,000,000 users concurrently. It must handle up to 50,000 courses with up to 5,000 users per course. Scholarwatch should accommodate up to 50 roles and 100 course categories, which may be nested up to 10 levels deep. These scalibility statistics are supported by moodle, which form the base of our LMS.

For details on performance and scalability of Moodle, refer to [**?**].

2.3.4 Compliance

The system must comply with educational data privacy laws, ensuring that all personal and academic data is handled securely and lawfully.

2.4 Domain Model

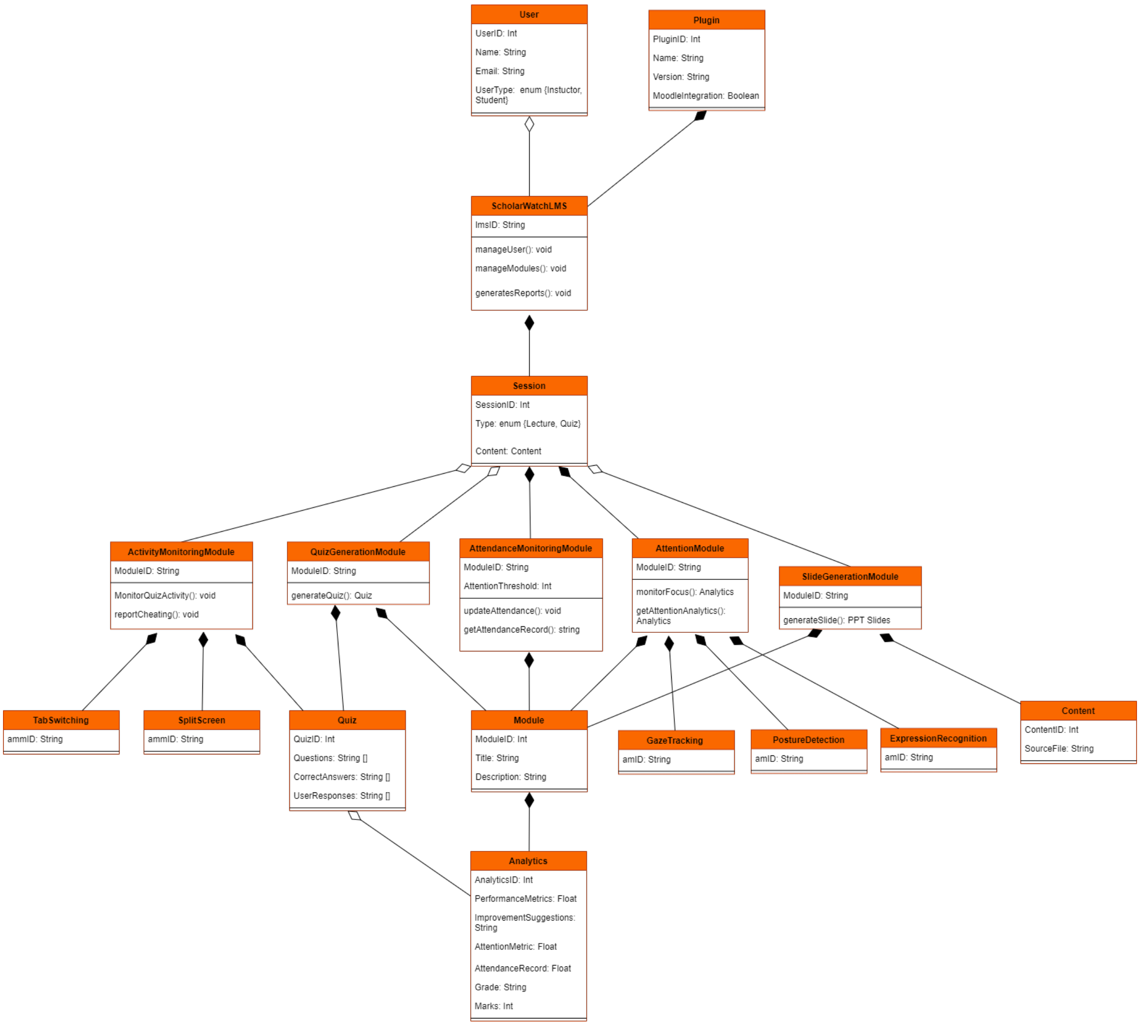

Figure 2.3: Domain model for ScholarWatch

Chapter 3

System Overview

Scholarwatch is an AI-driven Learning Management System (LMS) that integrates with Moodle to enhance teaching and learning. It provides instructors with insights into student engagement and performance, helping them tailor lessons effectively. For students, the platform monitors attention, tracks progress, and offers personalized content. With its intelligent tools, Scholarwatch simplifies course management and improves the overall learning experience.

3.1 Architectural Design

In our architecture diagram, various modules are represented as distinct layers, with their interactions illustrated through arrows. This layered architecture diagram showcases a modular design approach that allows for a clear separation of functionalities within the Scholarwatch system. Each layer corresponds to specific components, such as the user interface and content generation, highlighting how they communicate and collaborate to deliver an integrated learning experience.

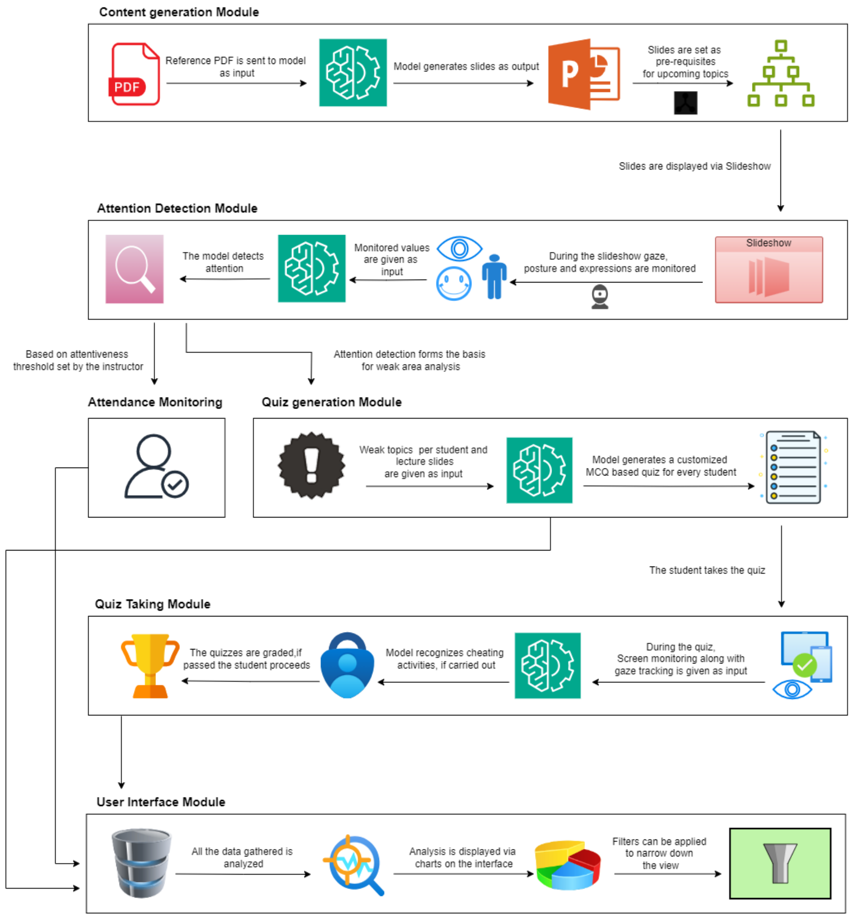

Figure 3.1: Architecture Diagram for ScholarWatch LMS

3.2 Design Models

Design Models for Procedural Approach

3.2.1 Activity Diagram

This activity diagram provides an overview of the typical workflow within Scholarwatch. Starting with the generation of slides, the system moves through different phases such as monitoring student attention, analyzing weak areas, and generating personalized quizzes. It demonstrates how the system continuously tracks student engagement, assesses quiz performance, and provides feedback to instructors.

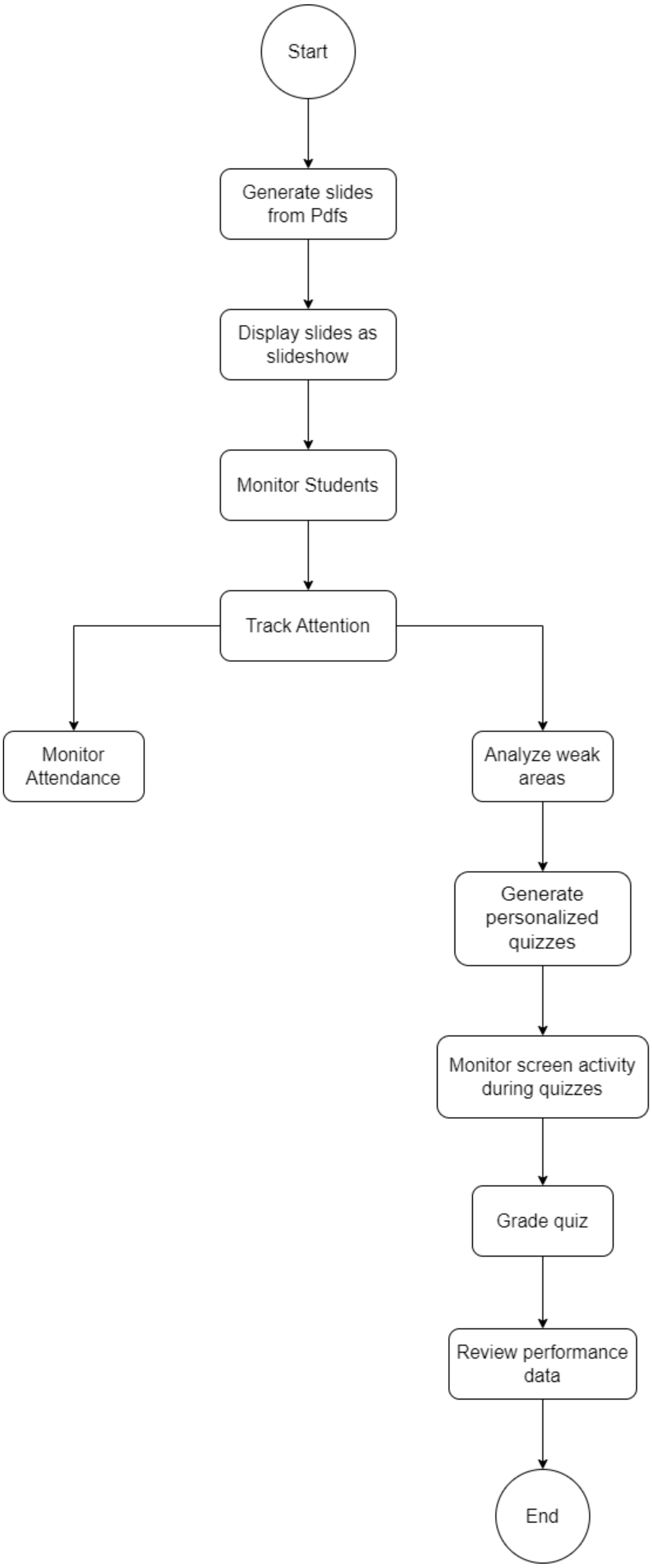

Figure 3.2: Activity Diagram for ScholarWatch

3.2.2 Data Flow Diagram

A Data Flow Diagram (DFD) for Scholarwatch visually represents how data moves within the system, illustrating interactions between different components. It highlights key processes such as slide generation, attendance monitoring, and quiz administration, detailing the inputs and outputs associated with each task. The DFD identifies data sources, such as instructors and students and aids in understanding the system's architecture and ensures efficient data management. Overall, the DFD serves as a valuable tool for stakeholders to comprehend and optimize the data flow within Scholarwatch.

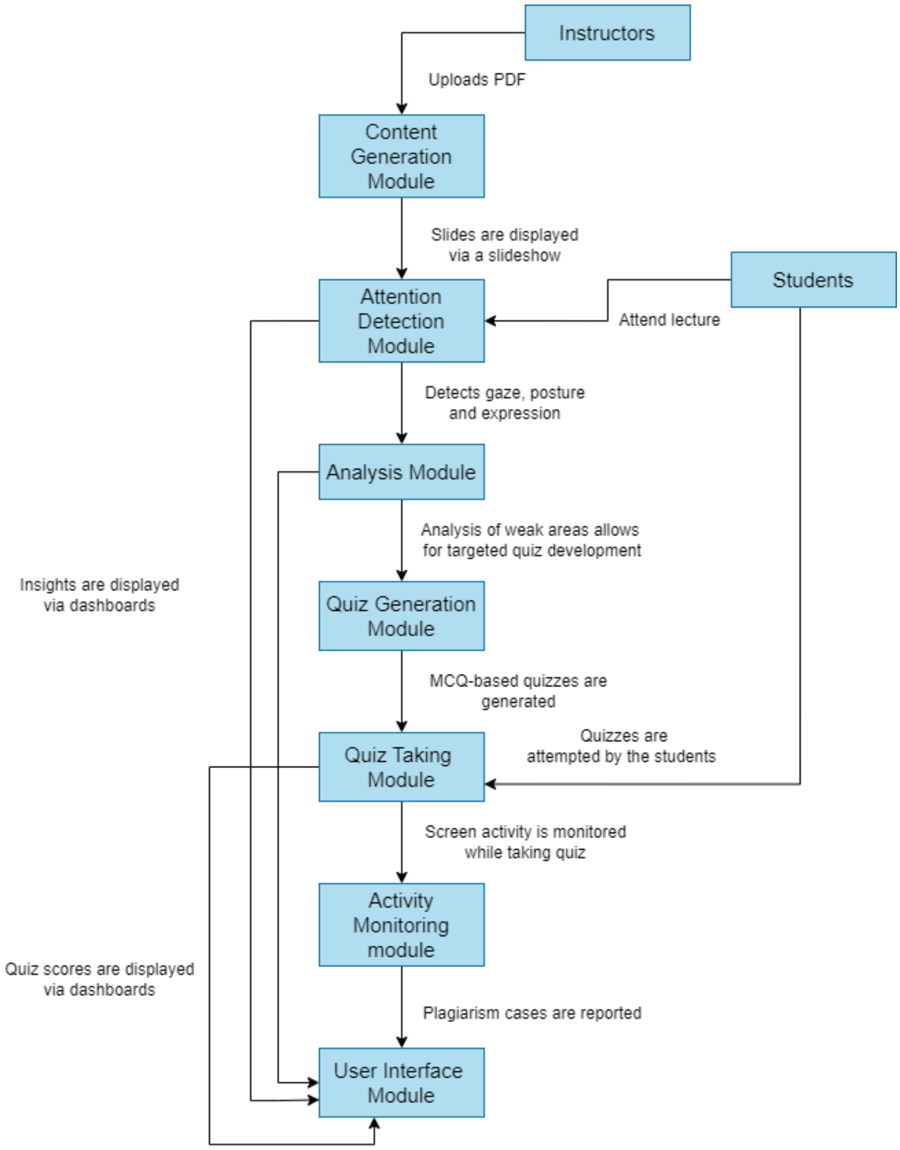

Figure 3.3: Data Flow Diagram for ScholarWatch

3.2.3 System Sequence Diagram

Instructor:

This diagram outlines how an instructor interacts with the LMS. Starting with creating a module and uploading materials, the system generates slides and tags prerequisites. The instructor can view class performance and use insights to refine lectures and quizzes. Key actions such as generation modules, generating personalized learning materials, and reviewing student performance are automated, helping the instructor better manage the course.

Student:

This diagram shows the typical sequence of actions for a student, beginning with enrolling in a module. The student views the lecture, with the system monitoring their attention and attendance. Quizzes are generated based on their engagement with the material, and activity during the quiz is tracked to ensure academic integrity. After the quiz, students can view their progress and performance, with visualizations helping them understand their standing in the course.

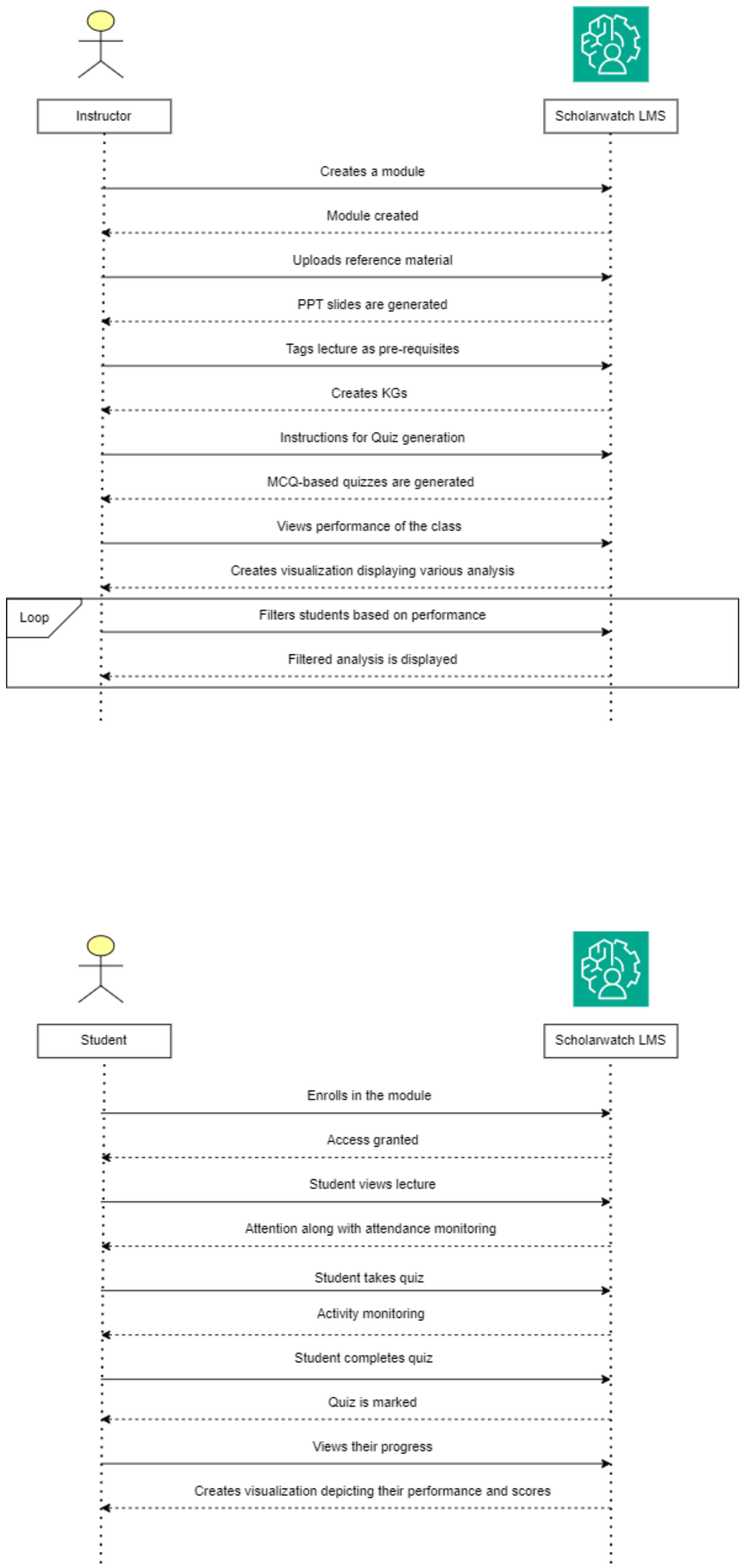

Figure 3.4: SSD for ScholarWatch

3.2.4 Sequence Diagram

The Sequence Diagrams for Scholarwatch depict the interactions between instructors, students, and the Learning Management System (LMS) throughout key processes. The instructor's sequence diagram illustrates the steps involved in creating a course, uploading materials, generating quizzes, and analyzing class performance, showcasing the system's response to each action.

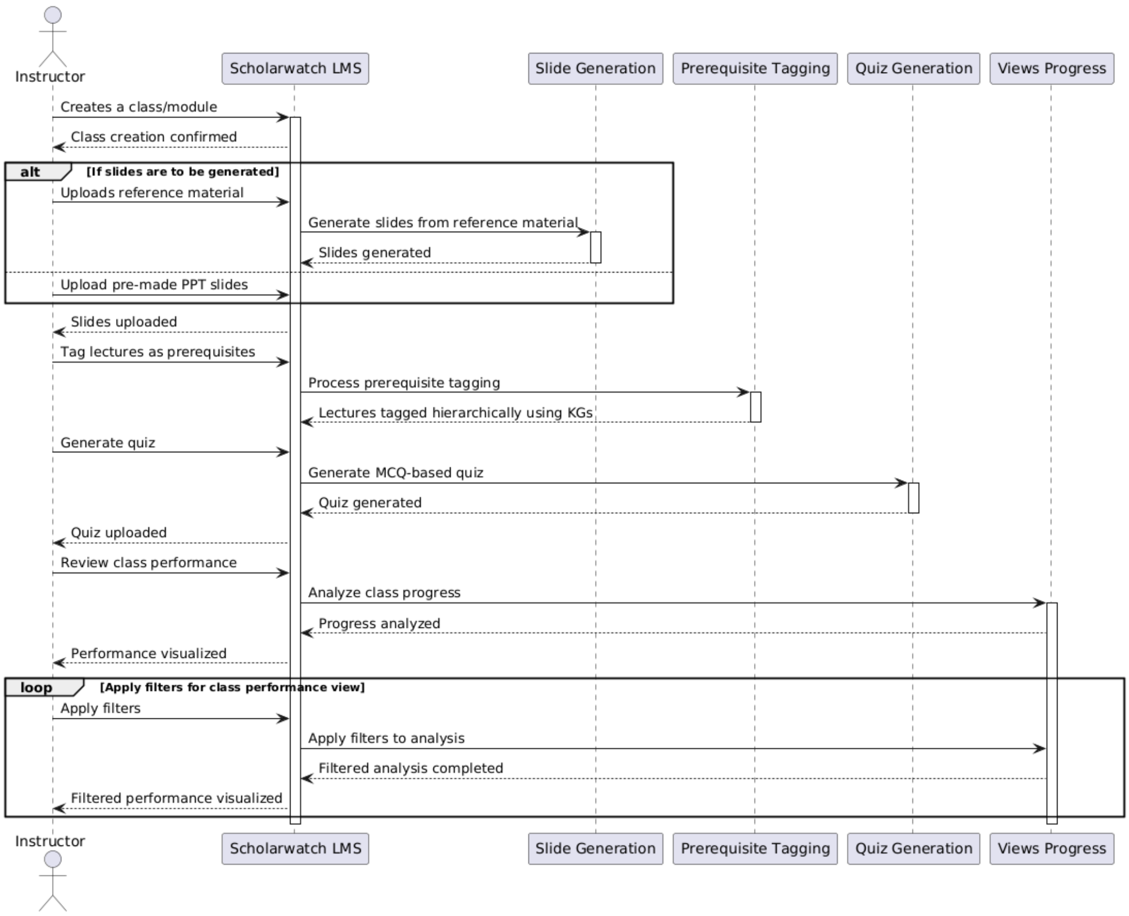

Figure 3.5: Sequence Diagram for Instructor

In contrast, the student's sequence diagram outlines the workflow for course enrollment, attending lectures, taking quizzes, and monitoring progress. Together, these diagrams provide a clear representation of the dynamic interactions within the Scholarwatch system, highlighting the efficiency and automation in managing educational activities. This clarity enhances the understanding of user roles and system functionalities, facilitating better design and implementation.

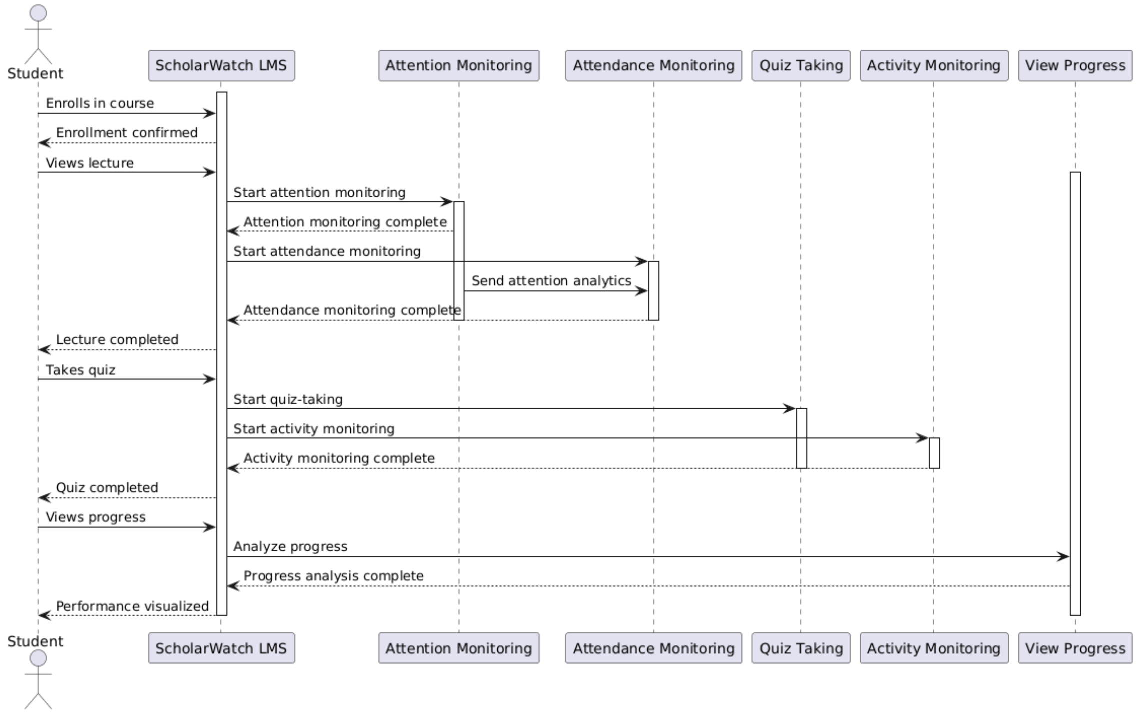

Figure 3.6: Sequence Diagram for Student

3.2.5 State Transition Diagram

The state transition diagram for Scholarwatch outlines the interactions and processes involved in the learning management system from both instructors' and students' perspectives. It begins with instructors uploading PDFs and creating content, leading to the display of slides during lectures. The Attention Detection Module actively monitors student attentiveness, detecting various attributes such as slouching, yawning, and confusion. These observations inform the Progress and Weak Area Analysis, which helps identify topics that require additional focus.

The Weak Area Analysis serves a dual purpose: it forms the basis for generating multiple-choice quizzes and provides valuable insights for instructors regarding content that needs further emphasis. As students attend lectures and take quizzes, their performance is evaluated, leading to either passing or failing outcomes. In cases of failure, review materials are assigned, and new quizzes can be generated for retakes. Finally, insights and progress are visually presented through a User Interface Module, ensuring both instructors and students have access to essential performance data.

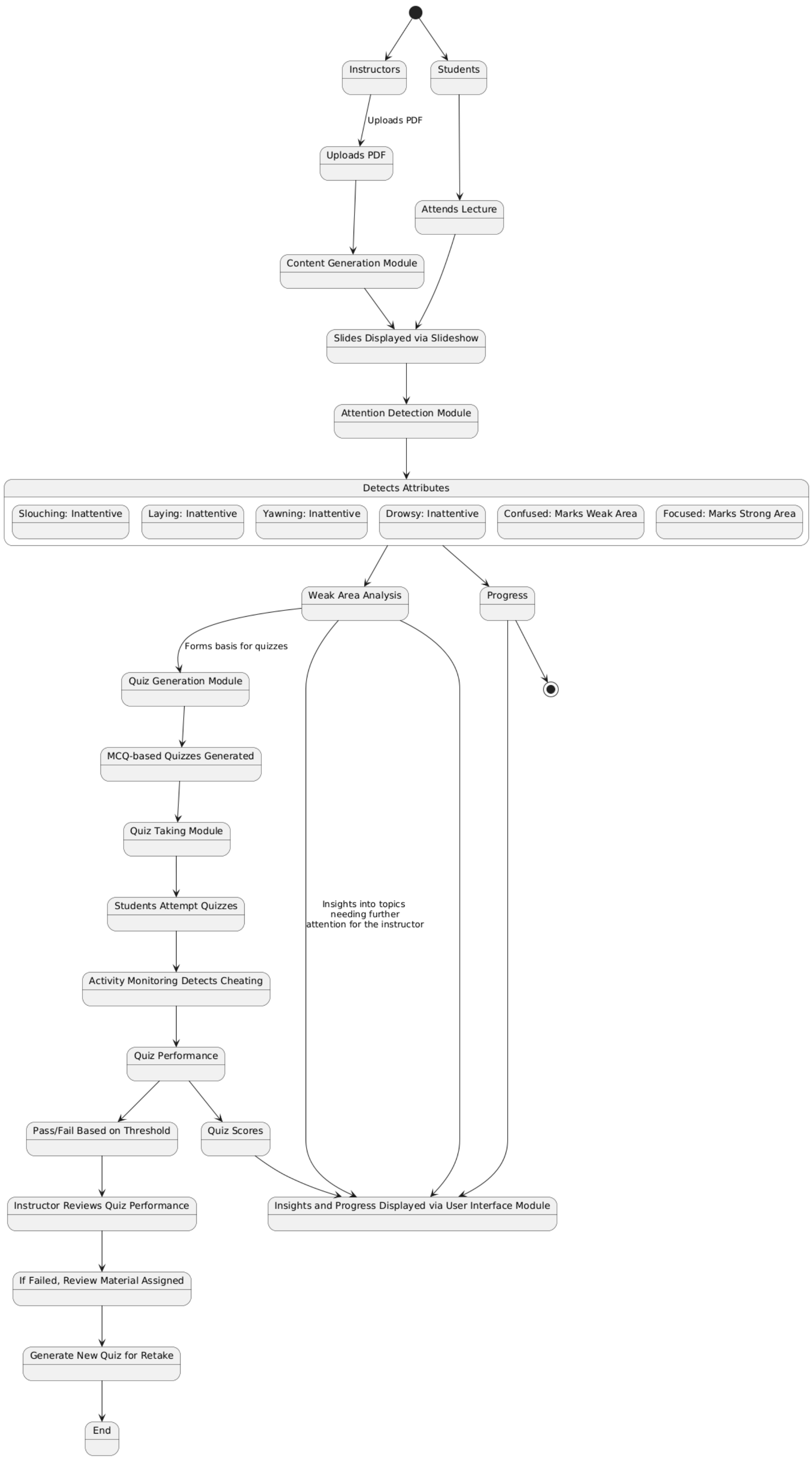

Figure 3.7: State Transition Diagram for ScholarWatch

3.3 Data Design

Moodle, at the time of writing this documentation, uses XMLDB as its database abstraction layer to provide a standard way to define and manage database structures using XML files. XMLDB allows interaction with a variety of databases such as MySQL, PostgreSQL, and MariaDB. It creates a database-neutral file that supports these and many more databases. [5].

Moodle consists of 487 tables and 4,424 columns. However, there are 20 core tables: Assignment, Analytics, Badges, Book, Chat, Choice, Course, Data, Enrolment, Feedback, Forum, Lesson, LTI, Page, Quiz, Scorm, Survey, Users, Workshop, and Wiki. [1][3].

Examulator.com, a website created and maintained by Marcus Green, a long-time Moodle developer, provides a platform for viewing and understanding the Moodle database. The database type used for schema generation was "MySQL - 8.0.36-0ubuntu0.22.04.1."

The ER diagram showing the relationships between all the tables can also be found at [2].

To insert or change data, Moodle makes use of APIs, which are essential when writing Moodle plugins. The APIs are divided into three categories: most-used general, other general, and activity module APIs. A few of the important APIs to note include the File API, which controls the storage of files in connection with various plugins, and the Data Manipulation API, which allows you to read and write to databases in a consistent and safe way. [4].

Bibliography

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